

Step-by-Step Guide to Create Backend Using XAMPP

◆ Step 1: Install XAMPP

1. Download XAMPP

- Visit: <https://www.apachefriends.org>
- Download latest version (PHP 8.x recommended)

2. Install

- Install in default path:
C:\xampp

3. Start Services

Open **XAMPP Control Panel** → Start:

- ☒ Apache
- ☒ MySQL

If both turn green → Server is running successfully.

◆ Step 2:

Case 1: Understand Folder Structure

All backend files must go inside:

C:\xampp\htdocs\

Create your project folder:

C:\xampp\htdocs\myproject

Example:

```
htdocs
├── myproject
│   ├── index.php
│   ├── config.php
│   ├── insert.php
│   └── fetch.php
```

Case 2: Shift Existing Project to XAMPP (If Already Developed)

If your website was created somewhere else like:

Desktop
Downloads
D Drive
VS Code folder

You MUST move it to **htdocs**.

Locate htdocs Folder

C:\xampp\htdocs\

This is Apache's root directory.

Copy Project into htdocs

Example:

Old location:

C:\Users\Swamy\Desktop\foodapp

Copy entire folder and paste inside:

C:\xampp\htdocs\

Now structure becomes:

C:\xampp\htdocs\foodapp

Do NOT paste files directly into htdocs. Always keep inside a folder.

Run the Project

Open in Browser

<http://localhost/foodapp/>

If [index.php](#) or [index.html](#) exists, project will open.

◆ Step 3: Create Database

1. Open phpMyAdmin

Go to browser:

<http://localhost/phpmyadmin>

2. Create Database

- Click **New**
- Database name: [myproject_db](#)
- Click Create

◆ Step 4: Create Table

Example: Users Table

```
CREATE TABLE users (  
  id INT AUTO_INCREMENT PRIMARY KEY,  
  name VARCHAR(100),  
  email VARCHAR(100),  
  phone VARCHAR(15),  
  created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP  
);
```

Click **SQL Tab** → **Paste** → **Run**

◆ Step 5: Create Database Connection (config.php)

Create file:

C:\xampp\htdocs\myproject\config.php

```
<?php
```

```
$servername = "localhost";
```

```
$username = "root";
```

```
$password = "";
```

```
$database = "myproject_db";
```

```
$conn = mysqli_connect($servername, $username, $password, $database);
```

```
if (!$conn) {
```

```
    die("Connection failed: " . mysqli_connect_error());
```

```
}
```

```
?>
```

◆ Step 6: Insert Data (Backend API)

Create file:

insert.php

```
<?php
```

```
include 'config.php';
```

```
if(isset($_POST['name']) && isset($_POST['email']) && isset($_POST['phone'])){

    $name = $_POST['name'];
    $email = $_POST['email'];
    $phone = $_POST['phone'];

    $sql = "INSERT INTO users (name, email, phone)
        VALUES ('$name', '$email', '$phone')";

    if(mysqli_query($conn, $sql)){
        echo "Data inserted successfully";
    } else {
        echo "Error: " . mysqli_error($conn);
    }
}
?>
```

◆ Step 7: Connect Frontend Form to Backend

Example `index.php`

```
<form action="insert.php" method="POST">
    <input type="text" name="name" placeholder="Enter Name" required>
    <input type="email" name="email" placeholder="Enter Email" required>
    <input type="text" name="phone" placeholder="Enter Phone" required>
    <button type="submit">Submit</button>
</form>
```

◆ Step 8: Test Your Backend

Open browser:

`http://localhost/myproject/`

Fill form → Submit → Check database table.

Go to:

phpMyAdmin → users table → Browse

◆ Step 9: Fetch Data (Optional API)

Create `fetch.php`

```
<?php
include 'config.php';

$sql = "SELECT * FROM users";
$result = mysqli_query($conn, $sql);

while($row = mysqli_fetch_assoc($result)){
    echo $row['name'] . " - " . $row['email'] . "<br>";
}
?>
```

Access:

<http://localhost/myproject/fetch.php>

◆ Step 10: Improve Security (Very Important)

Instead of direct query, use **Prepared Statements**:

```
$stmt = $conn->prepare("INSERT INTO users (name, email, phone) VALUES (?, ?, ?)");
$stmt->bind_param("sss", $name, $email, $phone);
$stmt->execute();
```

This prevents SQL Injection.

◆ Step 11: Project Structure (Professional Setup)

```
myproject/
├── config/
│   └── database.php
├── api/
│   ├── insert.php
│   └── fetch.php
├── assets/
│   ├── css/
│   └── js/
└── index.php
```

◆ Step 12: If You Want REST API (For Mobile App / React / Android)

Modify `insert.php`:

```
header("Content-Type: application/json");
```

```
$response = [];
```

```
if($stmt->execute()){
    $response["status"] = "success";
} else {
    $response["status"] = "error";
}
```

```
echo json_encode($response);
```

◆ Common Errors & Solutions

Error	Solution
Apache not starting	Port 80 busy → Change to 8080
MySQL not starting	Check port 3306
Access Denied	Username/password mismatch
Blank page	Enable error reporting

Enable errors:

```
error_reporting(E_ALL);  
ini_set('display_errors', 1);
```

ADDON on IMP Info

✗ Apache Not Starting

Reason: Port 80 busy

Solution:

- Change port to 8080
- Or stop Skype/IIS

Access:

<http://localhost:8080/projectname>

✗ MySQL Not Starting

Check:

- Port 3306 busy
 - Another MySQL installed
-

CSS / JS Not Loading

Wrong:

```
<link href="/css/style.css">
```

Correct:

```
<link href="css/style.css">
```

OR

```
<link href="/foodapp/css/style.css">
```

Database Connection Error

Check:

- Database name correct
 - MySQL running
 - Username/password correct
-

Professional Project Structure

After shifting, maintain clean structure:

```
htdocs
├── projectname
│   ├── config/
│   │   └── database.php
│   ├── api/
│   ├── assets/
│   │   ├── css/
│   │   ├── js/
│   │   └── images/
│   ├── index.php
│   └── .htaccess
```

Final Testing Checklist

Step	Status
XAMPP Installed	✓
Apache Running	✓
MySQL Running	✓
Project Copied to htdocs	✓
Database Created	✓
SQL Imported	✓
Config Updated	✓
Browser Tested	✓
Insert & Fetch Tested	✓